Energy calculations: revision worksheet

Year 8 SEAL Science · Chapter 7 Energy · Kai's Off-Grid Year · Lessons 1 to 7

Name: _____ Class: _____ Date: _____

18 questions · 69 marks · about 60 to 70 minutes · Extension: 7 bonus marks

How to use it. Work through Sections A to E in order. Each section starts with a worked example. Read it first, then try the questions.

Rules. Show your working each time. Give every final answer a unit. Use g = 10 N/kg. A calculator is allowed.

Difficulty labels. Easy: one step. Medium: two steps, a unit change, or checking working. Hard: rearrange a formula or link several ideas. Some questions show a level on each part.

Formula and units box

Quantity	Formula	Units and tips
Kinetic energy	**KE = ½mv²**	Mass in kg, speed in m/s, answer in joules (J). Square the speed first, then multiply by the mass, then halve it.
Gravitational potential energy	**GPE = mgh**	Mass in kg, g = 10 N/kg, height in m, answer in joules (J).
Energy chain	**GPE at the top = KE at the bottom**	Only when no energy is wasted. It works going up too: KE at the bottom = GPE at the top.
Efficiency	**Efficiency = useful energy out ÷ total energy in × 100%**	Never more than 100%. Wasted energy = total energy in − useful energy out.
Sankey diagram	**Arrow widths show energy amounts**	The input arrow is as wide as all the output arrows added together.
Energy used by an appliance	**Energy (Wh) = power (W) × time (h)**	1 kWh = 1,000 Wh.
Running cost	**Running cost = energy consumption (kWh) × tariff ($ per kWh)**	The tariff is the price of 1 kWh. Change cents to dollars first: 25 cents = $0.25.
Heat loss	**Heat loss after = heat loss before − total saving**	Add up the savings first. kWh per hour means the energy that leaks out each hour.
Rearranging (Hard parts)	**Find v: v² = KE ÷ (½ × m), then take the square root. Find h: h = GPE ÷ (m × g).**	Find time: time (h) = energy (Wh) ÷ power (W). Useful energy out = efficiency (as a decimal) × total energy in, so total energy in = useful energy out ÷ efficiency. In a chain, multiply the efficiencies: 0.6 × 0.5 = 0.3, so 30%.

Units: 1 kg = 1,000 g · 1 m = 100 cm · 1 kWh = 1,000 Wh · 1 kJ = 1,000 J · 100 cents = $1

How to set out a calculation

1 Formula	2 Convert	3 Substitute	4 Answer
Write the formula first.	Change units to kg, m, kWh or dollars.	Write the numbers into the formula.	Work it out. Write the unit.

Example: a 4 kg box slides at 3 m/s. KE = ½mv² → KE = ½ × 4 × 3² → KE = ½ × 4 × 9 → KE = 18 J

Rounding: if an answer has more than one decimal place, round it to 1 decimal place. Round money to the nearest cent.

Section	Skill	Marks
A	Kinetic energy	10
B	Gravitational potential energy and energy chains	11
C	Efficiency and Sankey diagrams	15
D	Power, energy and running cost	14
E	Mixed multi-step scenarios	19
Total		**69**

Section A: Kinetic energy (10 marks)

Calculate KE = ½mv², convert grams to kilograms, and rearrange to find a speed.

Worked example. A 30 kg wombat runs at 6 m/s. Calculate its kinetic energy.	
Formula	KE = ½mv²
Substitute	KE = ½ × 30 × 6²
Square first	6² = 36, so KE = ½ × 30 × 36
Multiply	KE = 540 J

A1 Red kangaroo **Easy**

A red kangaroo has a mass of 50 kg. It hops across a paddock at 7 m/s.

Calculate the kangaroo's kinetic energy. **2 marks**

Working

Answer: _____

A2 Fast bowler **Medium** (3 marks)

A cricket ball has a mass of 160 g. A fast bowler releases it at 35 m/s.

(a) Convert the mass of the ball to kilograms. **1 mark**

Working

Answer: _____

(b) Calculate the ball's kinetic energy as it leaves the bowler's hand. **2 marks**

Working

Answer: _____

A3 Spot the mistake: lawn bowls **Medium** (2 marks)

In lawn bowls, a player rolls a 1.6 kg bowl hard at 3 m/s. Tama works out its kinetic energy like this:

Tama's working

Line 1: KE = ½mv²
Line 2: KE = ½ × 1.6 × 3²
Line 3: KE = 0.8 × 6
Line 4: KE = 4.8 J

(a) State what Tama did wrong in line 3. **1 mark**

(b) Calculate the correct kinetic energy of the bowl. **1 mark**

Working

Answer: _____

A4 Toboggan Hard

A toboggan and its two riders have a total mass of 100 kg. Sliding across flat snow, the toboggan has 450 J of kinetic energy.

Calculate how fast the toboggan is moving. **3 marks**

Working

Answer: _____

Section B: Gravitational potential energy and energy chains (11 marks)

Calculate GPE = mgh, find the height gained, and follow energy from KE to GPE.

Worked example. An SES volunteer lifts an 18 kg sandbag onto a truck tray 150 cm above the ground. Calculate the GPE the sandbag gains.

Convert	150 cm = 1.5 m
Formula	GPE = mgh
Substitute	GPE = 18 × 10 × 1.5
Answer	GPE = 270 J

Energy chains. With no energy wasted, GPE at the top = KE at the bottom. Going up, it works the other way: KE at the bottom = GPE at the top.

B1 Lookout walk Easy

A bushwalker and her pack have a total mass of 70 kg. She climbs 150 m higher, from the car park to a lookout in Gariwerd (the Grampians).

Calculate the gravitational potential energy she gains. **2 marks**

Working

Answer: _____

B2 Chairlift **Medium** (3 marks)

A chairlift carries a skier and her gear (total mass 75 kg) up a mountain.

The bottom station is 1,450 m above sea level. The top station is 1,750 m above sea level.

(a) Calculate the height the skier rises. **1 mark**

Working

Answer: _____

(b) Calculate the gravitational potential energy the skier gains. **2 marks**

Working

Answer: _____

B3 Stomp rocket (6 marks)

A toy stomp rocket has a mass of 70 g. Stomping on its air pump launches it straight up at 20 m/s.

(a) **Medium** · Calculate the rocket's kinetic energy at launch. **2 marks**

Working

Answer: _____

(b) **Hard** · Assume no energy is wasted, so all of the kinetic energy becomes GPE at the top. Calculate the greatest height the rocket could reach. **2 marks**

Working

Answer: _____

(c) **Hard** · The rocket really reaches a height of 15 m, where it stops for an instant. Calculate how much energy was wasted on the way up. **2 marks**

Working

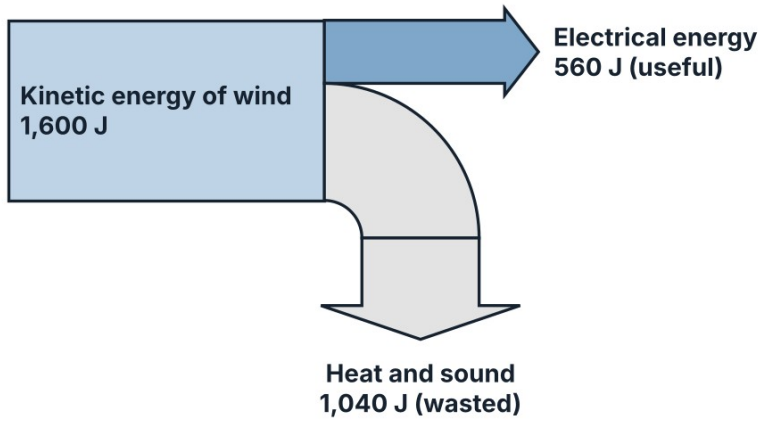
Answer: _____

Section C: Efficiency and Sankey diagrams (15 marks)

Calculate efficiency and wasted energy, read and complete Sankey diagrams, and compare two devices.

Worked example. Each second, the wind gives a small wind turbine at a weather station 1,600 J of kinetic energy. The turbine produces 560 J of electrical energy. Calculate its efficiency and the energy it wastes.

Formula	Efficiency = useful energy out ÷ total energy in × 100%
Substitute	Efficiency = 560 ÷ 1,600 × 100%
Answer	= 0.35 × 100% = 35%
Wasted	1,600 − 560 = 1,040 J, mostly heat and sound



C1 Electric chainsaw **Easy** (3 marks)

Each second, an electric chainsaw takes in 2,200 J of electrical energy. 1,430 J of it becomes kinetic energy of the moving chain.

(a) Calculate the efficiency of the chainsaw. **2 marks**

Working

Answer: _____

(b) Calculate the energy the chainsaw wastes each second. **1 mark**

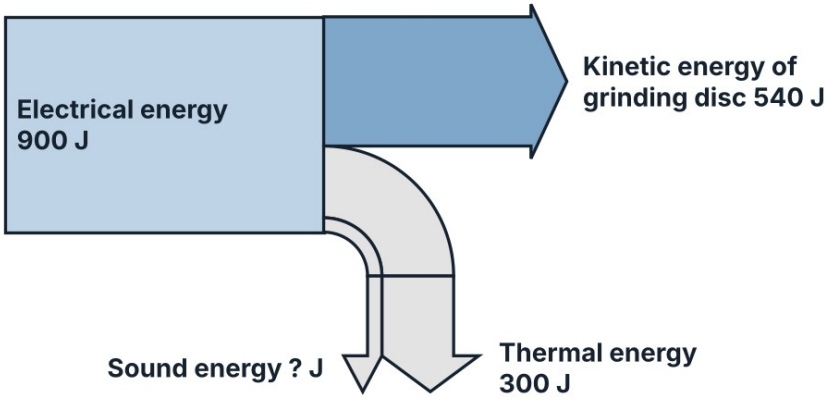
Working

Answer: _____

C2 Angle grinder **Medium** (4 marks)

The Sankey diagram shows what happens to the energy in a workshop angle grinder in one second. One value is missing.

Angle grinder, one second



(a) Calculate the missing amount of sound energy. **1 mark**

Working

Answer: _____

(b) Calculate the efficiency of the grinder. **2 marks**

Working

Answer: _____

(c) Heat spreads from the hot motor through the grinder's metal casing. Name this type of heat transfer. **1 mark**

Answer: _____

C3 Winch on a 4WD **Medium** (4 marks)

An electric winch pulls a 4WD that is stuck in mud on a forest track. Each second, the winch takes in 1,400 J of electrical energy. It is 50% efficient.

The useful energy is kinetic energy (the moving cable and 4WD). The wasted energy is thermal energy and sound. The thermal energy is 6 times the sound energy.

(a) Calculate the useful kinetic energy each second. **1 mark**

Working

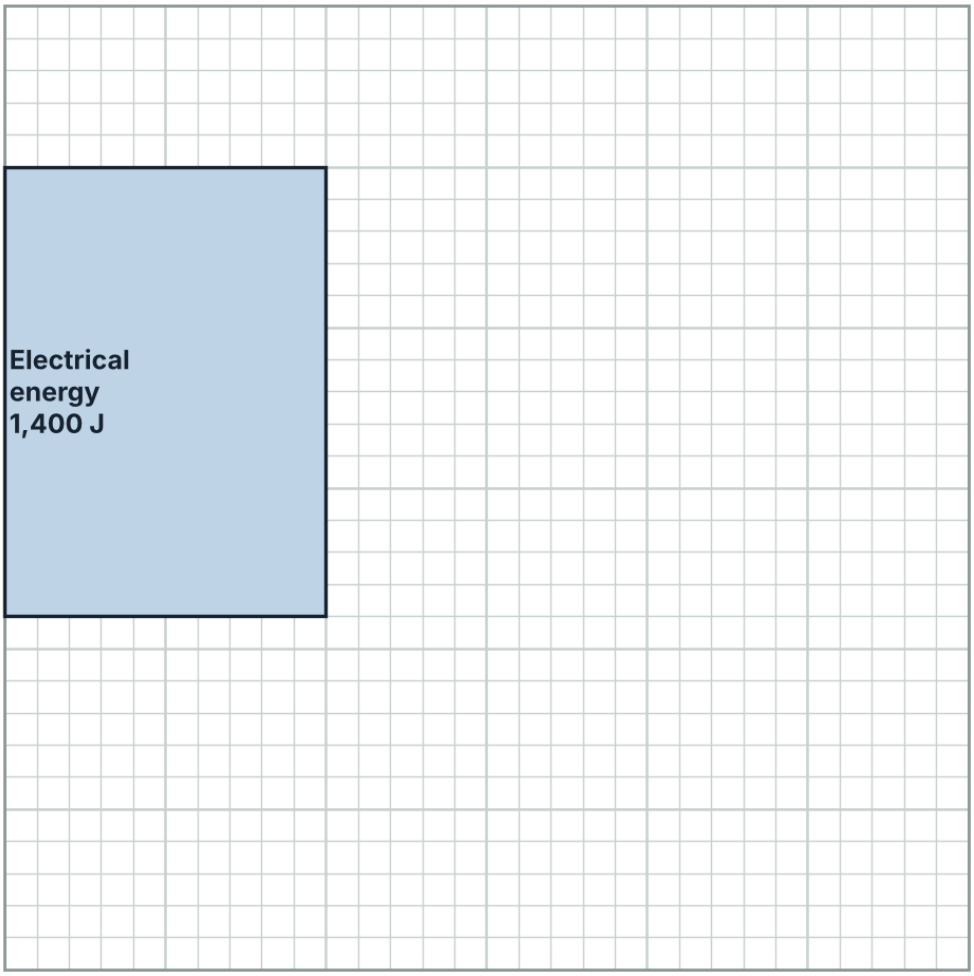
Answer: _____

(b) Work out the thermal and sound energy. Then complete the Sankey diagram on the grid. Start the useful arrow level with the top of the input arrow and draw it straight on to the right. Bend the wasted arrows down. Draw each arrow to scale and label it with its type and amount. A 1-square arrow can be a straight arrow pointing down, with its label beside it. **3 marks**

Working

Thermal energy: _____ Sound energy: _____

1 small square = 100 J



C4 Two workshop hoists (4 marks)

A mechanic has two electric hoists in her workshop.

Hoist A lifts a 250 kg engine 2 m up and out of a car. It uses 10,000 J of electrical energy.

Hoist B lifts a 180 kg gearbox 1.5 m up onto a stand. It uses 3,375 J of electrical energy.

(a) Easy · Calculate the useful energy for each lift: the GPE each load gains. **2 marks**

Working

Hoist A: _____

Hoist B: _____

(b) Hard · Calculate the efficiency of each hoist. Then state which hoist wastes the larger share of its energy input. **2 marks**

Working

Hoist A: _____

Hoist B: _____

Larger share wasted: _____

Section D: Power, energy and running cost (14 marks)

Use energy (Wh) = power (W) × time (h), change Wh to kWh, and work out running costs.

Worked example. An electric blanket uses 150 W. It is on for 4 hours a night, for 30 nights. Electricity costs 25 cents per kWh. Calculate the running cost.

Energy each night	Energy (Wh) = power (W) × time (h) = 150 × 4 = 600 Wh
Convert	600 Wh ÷ 1,000 = 0.6 kWh
For 30 nights	0.6 × 30 = 18 kWh
Running cost	kWh × tariff = 18 × $0.25 = $4.50

D1 Boot dryer Easy (3 marks)

A boot-drying rack in a ski lodge uses 350 W. It runs for 8 hours overnight. Electricity costs 35 cents per kWh.

(a) Calculate the energy the rack uses overnight, in Wh. **1 mark**

Working

Answer: _____

(b) Convert your answer to kWh. **1 mark**

Working

Answer: _____

(c) Calculate the cost of running the rack for one night. **1 mark**

Working

Answer: _____

D2 Two pool pumps **Medium** (4 marks)

A swimming club is choosing between two pumps for its outdoor pool. Both keep the water equally clean.

Pump 1 uses 1,600 W and must run for 6 hours a day. Pump 2 uses 550 W and must run for 12 hours a day.

(a) Calculate the energy each pump uses per day, in kWh. **2 marks**

Working

Pump 1 (per day): _____

Pump 2 (per day): _____

(b) Electricity costs 25 cents per kWh. Calculate the cost of running each pump for one week (7 days). **2 marks**

Working

Pump 1 (per week): _____

Pump 2 (per week): _____

D3 Dehumidifiers (4 marks)

A ski lodge needs a dehumidifier for its gear-drying room. The energy labels of two models show:

Model	Energy consumption (kWh per year)
X	350
Y	245

Electricity costs 26 cents per kWh.

(a) **Medium** · Calculate the yearly running cost of each model. Use the table. **2 marks**

Working

Model X: _____

Model Y: _____

(b) Hard · Calculate how much Model Y saves each year, as a percentage of Model X's running cost. **2 marks**

Working

Answer: _____

D4 Air compressor (3 marks)

A workshop air compressor uses 1,250 W. The workshop pays an off-peak tariff of 20 cents per kWh.

(a) Medium · Calculate how many kWh of electrical energy $1 buys. **1 mark**

Working

Answer: _____

(b) Hard · Calculate how long the compressor can run on $1 of electricity. **2 marks**

Working

Answer: _____

Section E: Mixed multi-step scenarios (19 marks)

Link the calculations to heat transfer (Lesson 3), friction (Lesson 5) and heat leaks in buildings (Lesson 7).

No worked example here. Each question uses skills from Sections A to D. Plan your steps before you start.

E1 Cable car to a lookout (7 marks)

A cable car and its passengers have a total mass of 900 kg. An electric motor pulls the car 150 m higher, up to a mountain lookout. The motor system is 75% efficient.

Tip: you can write large answers in kJ (1 kJ = 1,000 J).

(a) Easy · Calculate the GPE the car and passengers gain on the trip up. **2 marks**

Working

Answer: _____

(b) Hard · The GPE is the useful energy. Calculate the electrical energy the motor system uses for the trip. **2 marks**

Working

Answer: _____

(c) Medium · Calculate the energy wasted on the trip. **1 mark**

Working

Answer: _____

(d) Hard · On the next trip, the passengers are lighter, so the car and passengers have a total mass of 600 kg. The climb is still 150 m and the motor system is still 75% efficient. Calculate the electrical energy it uses for this trip. **2 marks**

Working

Answer: _____

E2 Brakes on a mountain road (6 marks)

A car and its passengers have a total mass of 1,200 kg. Coming down a mountain road, the driver brakes from 15 m/s to a stop before a sharp bend.

Ignore the change in height. Assume all of the car's kinetic energy ends up as thermal energy in its four brake discs.

(a) Easy · Calculate the car's kinetic energy at 15 m/s. **2 marks**

Working

Answer: _____

(b) Easy · Name the force that acts where the brake pads rub on the brake disc. **1 mark**

Answer: _____

(c) Medium · Assume the four brake discs share the thermal energy equally. Calculate the thermal energy each disc gains. **1 mark**

Working

Answer: _____

(d) Medium · Each metal brake disc has vents: gaps that air flows through as the wheel turns. Explain how the vents help a hot disc cool down. **2 marks**

E3 Is this claim correct? Ski lodge bunk room (6 marks)

On a winter night, a ski lodge's bunk room loses 3.0 kWh of heat each hour.

The manager insulates under the floor (R2.0) in two stages. Insulating the first half of the floor cuts the heat loss by 0.45 kWh per hour. Insulating the second half cuts it by another 0.45 kWh per hour.

(a) Easy · Calculate the room's heat loss per hour after both stages. **2 marks**

Working

Answer: _____

(b) Hard · The manager says: “Insulating the floor has cut the room's heat loss in half.” Is she correct? Use a calculation to decide. **2 marks**

Working

Verdict: _____

(c) Medium · Calculate how much less heat the room loses over a 12-hour night after both stages. **1 mark**

Working

Answer: _____

(d) Easy · The floor insulation also comes as R4.0. State which one, R2.0 or R4.0, slows heat flow more. **1 mark**

Answer: _____

Extension: for early finishers (bonus marks, not in the total)

X1 Medical delivery drone Hard

A 2 kg delivery drone takes medicine to a remote farm. It climbs 90 m, then flies level at 15 m/s. To do this, its battery supplies 6,750 J of electrical energy.

(a) Calculate the drone's GPE and KE when it is 90 m up and flying at 15 m/s. Add them to find its total useful energy. **2 marks**

Working

Answer: _____

(b) Calculate the efficiency of the drone for this trip. **1 mark**

Working

Answer: _____

(c) Name one form the wasted energy takes. **1 mark**

Answer: _____

X2 Solar-powered radio Hard

At a remote ranger station, a solar panel charges a battery that runs the two-way radio. The panel is 18% efficient. The battery gives back 85% of the energy put into it.

(a) Calculate the percentage of the sunlight's energy that reaches the radio as electrical energy. **1 mark**

Working

Answer: _____

(b) The rangers can afford one change: a 24% panel, or a battery that gives back 95%. Which change gives the greater overall efficiency? Show calculations. **2 marks**

Working

Better change: _____

My score: _____ / 69 Bonus: _____ / 7

Check your answers with the worked solutions. For each mistake, write which step went wrong: formula, units, substitution or arithmetic.